Analysis group meeting

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List of things done

- The cooking for the DC calibration is done.
- Vz sector independent fits for pion plus/minus for Sn, Cu, and C targets.
- Extraction of the mean and sigma for Vx, Vy, and Vz for all targets
- Extract nuclear transparency vs. Q2 and lc.
- Process files (pass0v7) using the yaml file from Veronique to produce the detached vertex bank.
- Try to improve K^0 identification with detached vertex but fail
- ALERT: Check how often a hit from GEMC comes from a secondary particle, and check particle PID and energy to understand the impact of these particles.

To do list

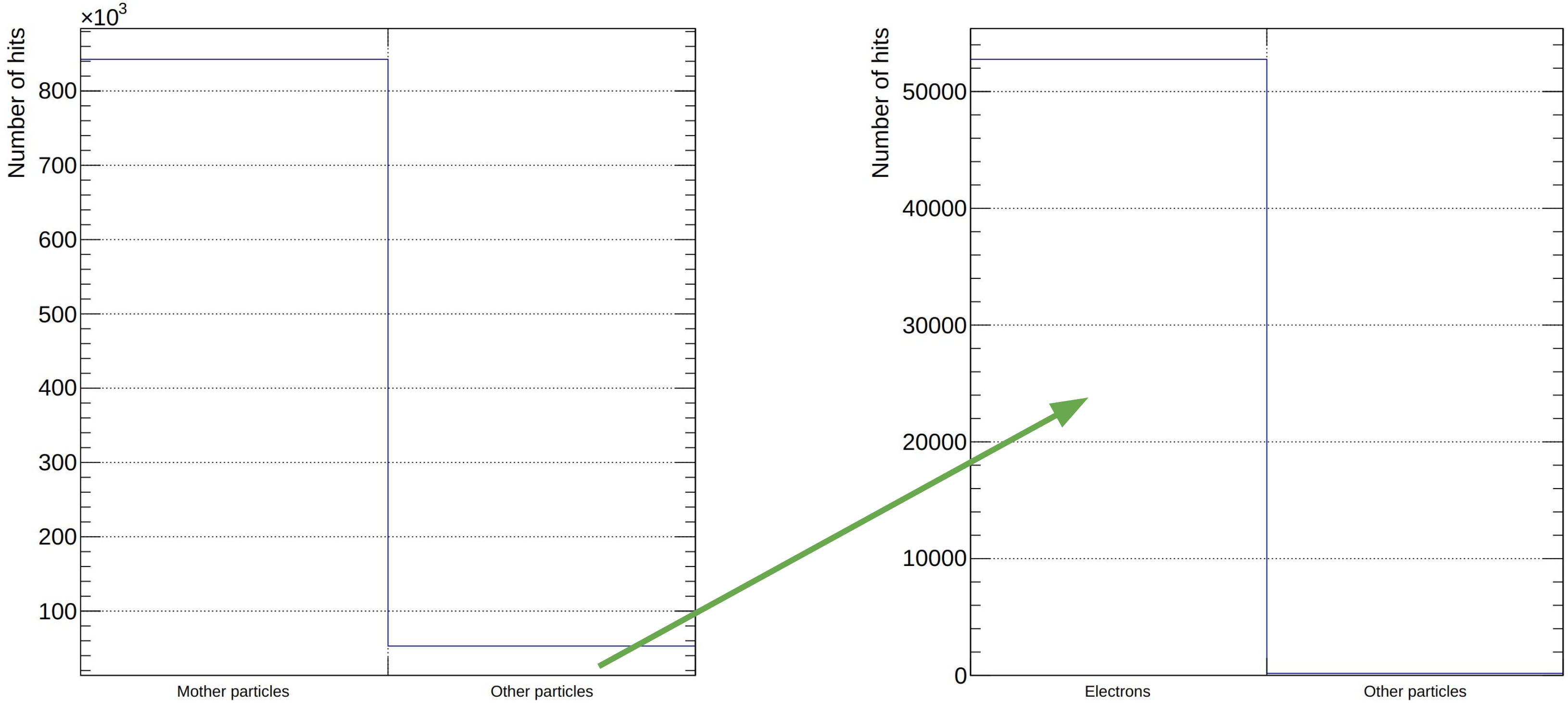
- RG-D study:
  - Cross check with Matt the Vx, Vy and Vz mean and sigma for each targets
  - Cross check for the nuclear transparency vs Q^2
  - Work on K^0 analysis:
    - Improve K^0 identification with detached vertex
    - Compute multiplicity ratio and asymmetry

- AI for ALERT
  - Redo the efficiency/purity analysis using luminosity simulation to have a better background (have to redo the simulation, volatile have delete the files)
  - Comparison between conventional and AI track finding

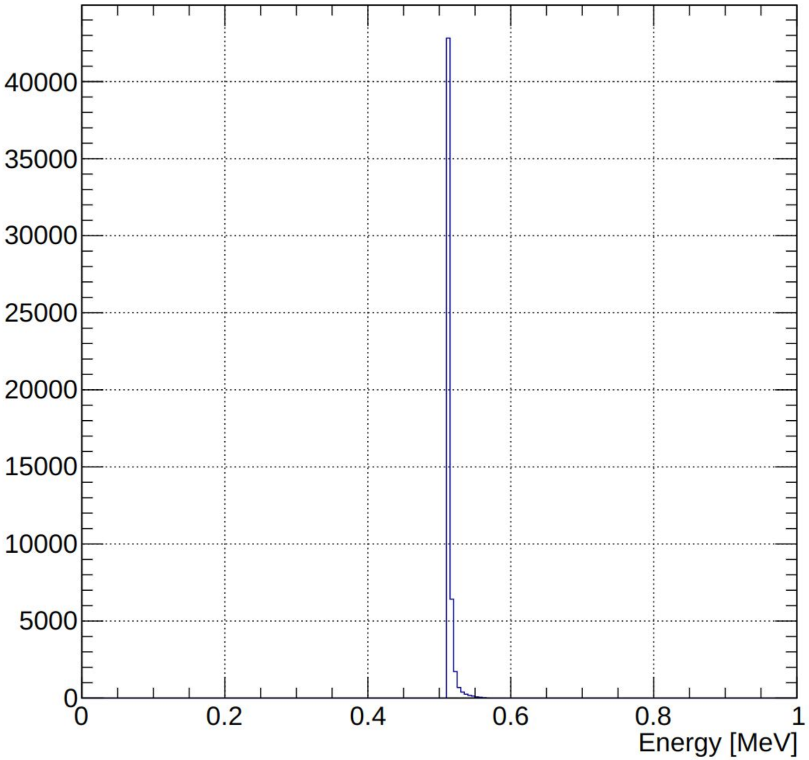
Backup

Study secondary particles

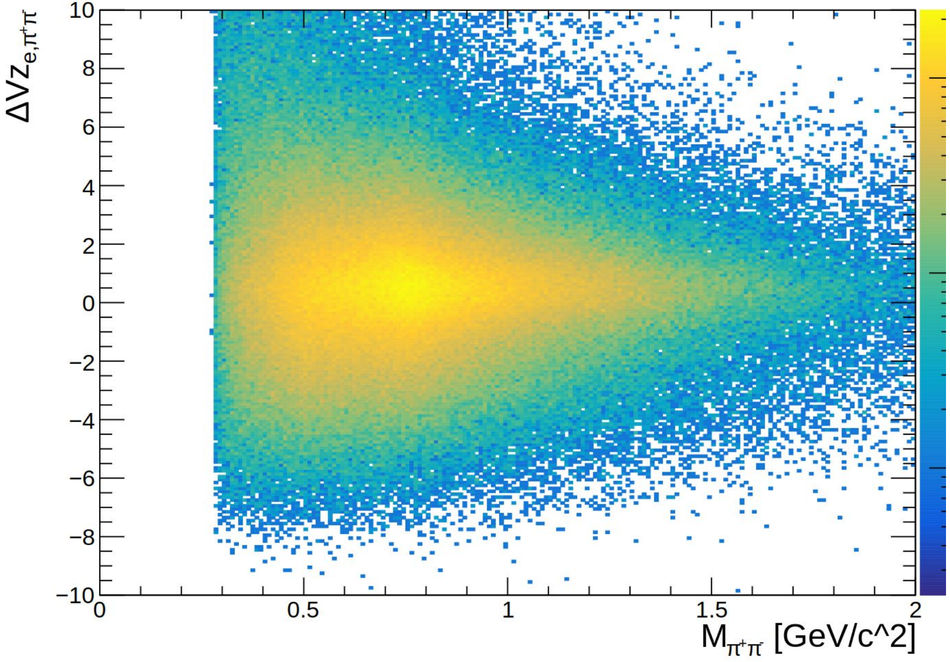
Determine the frequency of secondary particle production and its ability to cause hits in the AHDC.
Data use: 1 proton p∈ [60, 250] MeV/c, φ ∈ [0, 360]°, θ ∈ [60, 120]° and vz ∈ [-15, 15] cm.

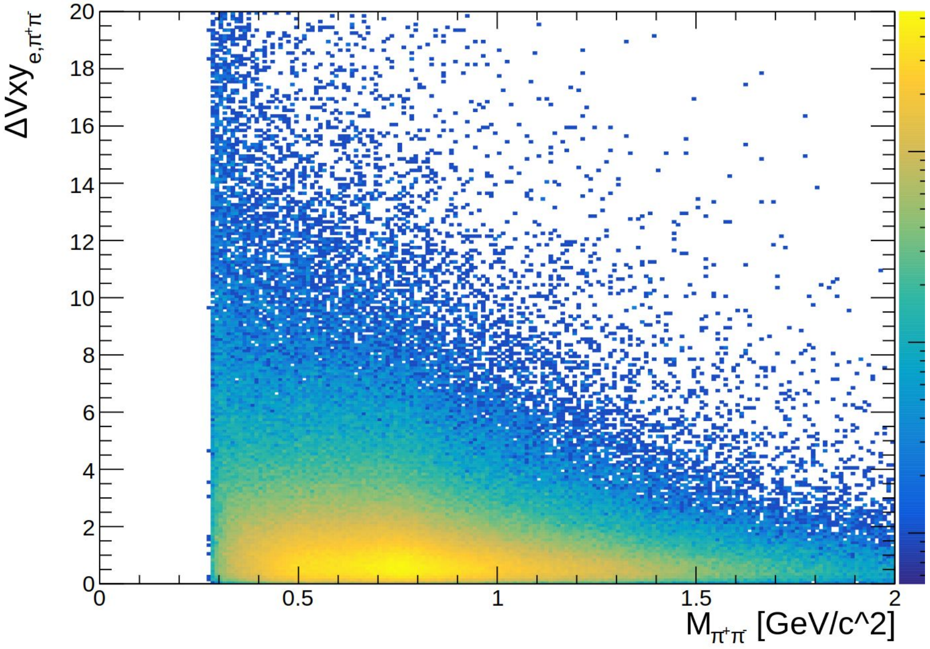


Study secondary particles



K^0 study


$\Delta Vz_{e,\pi^+\pi^-}$
10
8
6
4
2
0
−2
−4
−6
−8
−10
0
0.5
1
1.5
2
$M_{\pi^+\pi^-}$ [GeV/c^2]


$\Delta Vxy_{e,\pi^+\pi^-}$
20
18
16
14
12
10
8
6
4
2
0
0
0.5
1
1.5
2
$M_{\pi^+\pi^-}$ [GeV/c^2]